

FastBoot

Core logic: Select the first bootable device, jump other device scan

```
UINTN i, count;
LoadOptions = EfiBootManagerGetLoadOptions (&count, LoadOptionTypeBoot);
DEBUG ((DEBUG_ERROR, "zjdbg enter FastBoot mode, count=%d\n", count));
for (i=0; i<count; i++) {
    DEBUG ((DEBUG_ERROR, "zjdbg no:%d boot description:%s\n", i, LoadOptions[i].Description));
    DEBUG ((DEBUG_ERROR, "zjdbg Attributes:0x%x LOAD_OPTION_ACTIVE:0x%x\n", LoadOptions[i].Attributes, LOAD_OPTION_ACTIVE));
    if (LoadOptions[i].Attributes & LOAD_OPTION_ACTIVE) {
        PrintBootTime("Setup Done");
        EfiBootManagerBoot(&LoadOptions[i]);
        // CpuDeadLoop();
    }
}
```

`LoadOptions = EfiBootManagerGetLoadOptions (&count, LoadOptionTypeBoot);`

→ Return the array of startup devices

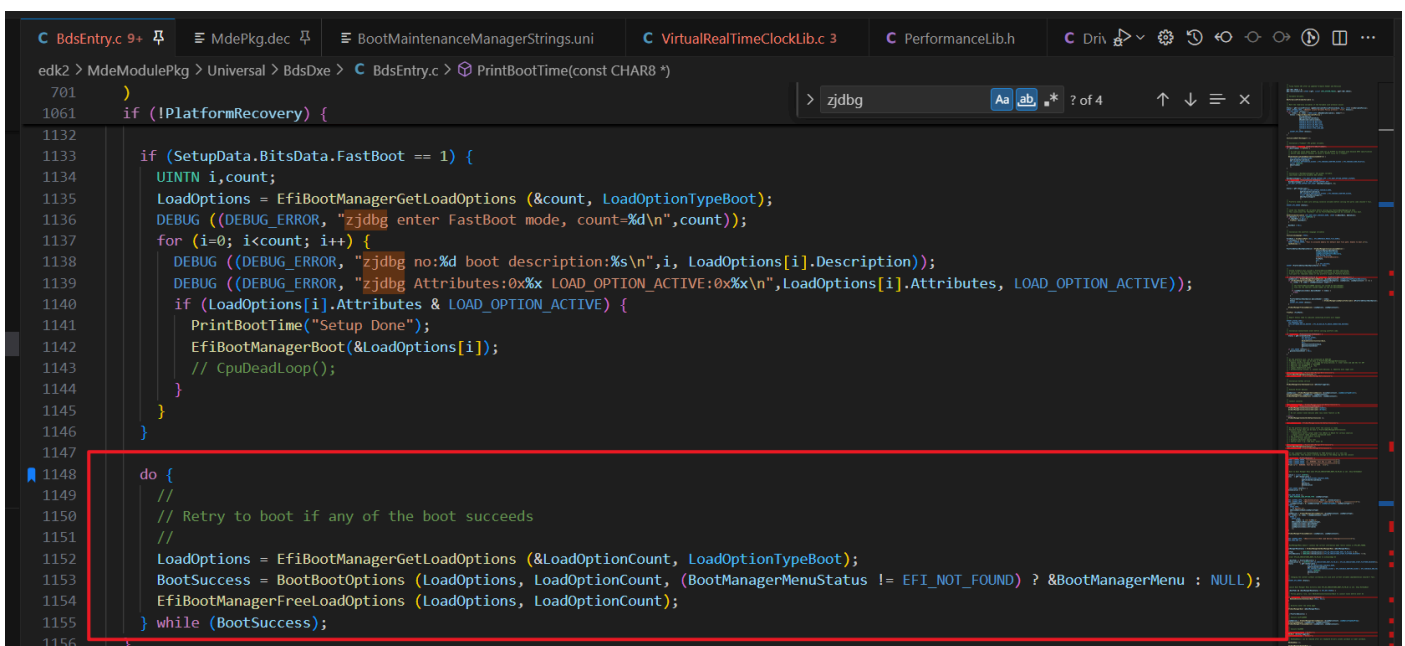
`LoadOptions[i].Attributes & LOAD_OPTION_ACTIVE`

→ Select the first bootable device

`EfiBootManagerBoot(&LoadOptions[i]);`

→ Activate the selected device

Origin logic



```
BdsEntry.c 9+
MdePkg.dec
BootMaintenanceManagerStrings.uni
VirtualRealTimeClockLib.c 3
PerformanceLib.h
Dri

edk2 > MdeModulePkg > Universal > BdsDxe > C BdsEntry.c > PrintBootTime(const CHAR8 *)
701
1061 if (!PlatformRecovery) {
1132
1133     if (SetupData.BitsData.FastBoot == 1) {
1134         UINTN i, count;
1135         LoadOptions = EfiBootManagerGetLoadOptions (&count, LoadOptionTypeBoot);
1136         DEBUG ((DEBUG_ERROR, "zjdbg enter FastBoot mode, count=%d\n", count));
1137         for (i=0; i<count; i++) {
1138             DEBUG ((DEBUG_ERROR, "zjdbg no:%d boot description:%s\n", i, LoadOptions[i].Description));
1139             DEBUG ((DEBUG_ERROR, "zjdbg Attributes:0x%x LOAD_OPTION_ACTIVE:0x%x\n", LoadOptions[i].Attributes, LOAD_OPTION_ACTIVE));
1140             if (LoadOptions[i].Attributes & LOAD_OPTION_ACTIVE) {
1141                 PrintBootTime("Setup Done");
1142                 EfiBootManagerBoot(&LoadOptions[i]);
1143                 // CpuDeadLoop();
1144             }
1145         }
1146     }
1147
1148     do {
1149         //
1150         // Retry to boot if any of the boot succeeds
1151         //
1152         LoadOptions = EfiBootManagerGetLoadOptions (&LoadOptionCount, LoadOptionTypeBoot);
1153         BootSuccess = BootBootOptions (LoadOptions, LoadOptionCount, (BootManagerMenuStatus != EFI_NOT_FOUND) ? &BootManagerMenu : NULL);
1154         EfiBootManagerFreeLoadOptions (LoadOptions, LoadOptionCount);
1155     } while (!BootSuccess);
1156 }
```

```

VOID EFIAPI PrintBootTime(CONST CHAR8 *Stage)
{
    UINT64 Now = GetPerformanceCounter();
    UINT64 Freq = GetPerformanceCounterProperties(NULL, NULL);

    if (gBootStart == 0) {
        gBootStart = Now;
        DEBUG((DEBUG_ERROR, "[BOOT TIME] Start\n"));
        return;
    }

    UINT64 Us = (Now - gBootStart) * 1000000 / Freq;
    DEBUG((DEBUG_ERROR, "[BOOT TIME] %a: %ld us (%ld.%03ld s)\n",
    Stage, Us, Us / 1000000, (Us % 1000000) / 1000));
    gBootStart = Now;
}

```

This function is used to print boot time.

GetPerformanceCounter();

→ return tick number

GetPerformanceCounterProperties(NULL, NULL);

→ return the frequency

tick number / frequency = time (unit: s)

Rendering

Form to Show Questions with union and bit UarStore

Nested BIT fields in efivarstore

Fast Boot Help

Fast Boot ☒ <Disable>
NEST_BIT check box ☒
NEST_BIT one-of <LargeBootList>
NEST_BIT numeric [2]
Use byte field in <LargeBootList>
NEST_BIT structure

BIT fields in efivarstore

BIT check box ☒
BIT numeric [4]
BIT one-of <LargeBootList>

Nested BIT fields in bufferstore

NEST_BIT check box ☒



↑↓=Move Highlight

F9=Reset to Defaults
<Enter>=Select Entry

F10=Save
Esc=Exit